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EDUCATION

Ph.D., Physics, University of California, Berkeley Expected May 2027

Advisor: Prof. William Holzapfel

B.A., Physics, *cum laude*, Barnard College, Columbia University December 2019

Advisors: Prof. Charles Hailey, Dr. Kaya Mori

RESEARCH EXPERIENCE

Graduate Student Researcher and NSF Graduate Research Fellow, University of California, Berkeley August 2020 – Present

South Pole Telescope: SPT-3G and SPT-3G+ experiments

- Developed a complete characterization of mm-wave polarized atmospheric emission at the South Pole to mitigate large-scale polarization systematics for SPT-3G (Coerver et al. 2025, *ApJ*)
- Placed a new constraint on anisotropic cosmic birefringence using SPT-3G data (Coerver et al., in prep)
- Set up and operated a cryogenic testbed for SPT-3G+ detector and readout electronics R&D.

Undergraduate Researcher, Columbia University, New York June 2017 – December 2019

NuSTAR experiment

- Conducted a multiwavelength investigation of the pulsar wind nebula DA 495 combining HAWC, VERITAS, and NuSTAR data (Coerver et al. 2019, *ApJ*)
- Analyzed NuSTAR and Chandra observations of new X-ray transients in the Galactic Center region (Mori et al., incl. A. Coerver, 2019, *ApJ*)

NSF CERN REU Summer Intern, CERN, Geneva, Switzerland June – August 2019

SHINE Experiment, Experimental Physics Department

DOE SULI Summer Intern, Lawrence Berkeley National Laboratory, Berkeley, CA June – August 2018

CMB Group, Physics Division

FELLOWSHIPS & AWARDS

NSF Graduate Research Fellow

August 2020 – August 2025

Henry A. Boorse Prize in Physics, Barnard College

December 2019

Junior Phi Beta Kappa, Barnard College

December 2019

Summer Research Institute Fellow, Barnard College

May 2017

FIRST-AUTHOR & SELECTED PUBLICATIONS

Coerver, A. et al. “Searching for Anisotropic Cosmic Birefringence with SPT-3G”. In preparation. 2026.

Natoli, T., et al. (incl. **A. Coerver**). *SPT-3G+: A Cosmic Microwave Background Experiment for the South Pole Telescope*. 2026. arXiv: [2608.20236 \[astro-ph.CO\]](#).

Coerver, A. et al. “Measurement and Modeling of Polarized Atmosphere at the South Pole with SPT-3G”. *The Astrophysical Journal* (2025). DOI: [10.3847/1538-4357/ada35d](#). arXiv: [2407.20579 \[astro-ph.IM\]](#).

Zebrowski, J. A., et al. (incl. **A. Coerver**). “SPT-3G D1: Constraints on inflationary gravitational waves with two years of SPT-3G data”. *Physical Review D* (2025). DOI: [10.1103/33wm-c6pt](#). arXiv: [2505.02827 \[astro-ph.CO\]](#).

T. de Haan, A. Suzuki, S. T. P. Boyd, R. H. Cantor, **Coerver, A.**, M. A. Dobbs, R. Hennings-Yeomans, W. L. Holzapfel, A. T. Lee, G. I. Noble, G. Smecher, and J. Zhou. “Recent Advances in Frequency-Multiplexed TES Readout: Vastly Reduced Parasitics and an Increase in Multiplexing Factor with Sub-Kelvin SQUIDS”. *Journal of Low Temperature Physics* (2020). DOI: [10.1007/s10909-020-02403-8](#). arXiv: [1908.07642 \[astro-ph.IM\]](#).

Coerver, A., P. Wilcox, H. Zhang, B. L. Dingus, E. V. Gotthelf, C. J. Hailey, T. B. Humensky, P. Kaaret, H. Li, K. Mori, R. Mukherjee, N. Park, and H. Zhou. “Multiwavelength Investigation of Pulsar Wind Nebula DA 495 with HAWC, VERITAS, and NuSTAR”. *The Astrophysical Journal* (2019). DOI: [10.3847/1538-4357/ab21d0](#). arXiv: [1905.07327 \[astro-ph.HE\]](#).

INVITED & CONTRIBUTED TALKS

* denotes invited talk

New Frontiers in Cosmology, A Coruña, Spain

August 2026

A New Constraint on Anisotropic Birefringence with SPT-3G

60th Rencontres de Moriond Cosmology, La Thuile, Italy

April 2026

A New Constraint on Anisotropic Birefringence with SPT-3G

*“Cosmmute” Talk, SLAC, Palo Alto, CA

October 2025

Constraining Anisotropic Cosmic Birefringence with SPT-3G

APS Global Physics Summit, Anaheim, CA

March 2025

SPT-3G: Measurement and Mitigation of mm-wave Polarized Atmospheric Emission

*CMB-S4 Special Session, URSI National Radio Science Meeting, Boulder, CO	January 2025
<i>Polarized Atmospheric Properties and Modeling for CMB-S4</i>	
*Astrophysics Colloquium, University of Melbourne	September 2024
<i>Measurement and Modeling of Polarized Atmosphere at the South Pole</i>	
*Plenary Speaker, CMB-S4 Workshop, UIUC, Urbana-Champaign, IL	July 2024
<i>Measurement and Modeling of Polarized Atmosphere</i>	
American Astronomical Society Meeting, Seattle, WA	January 2018
<i>DA 495: An Aging Pulsar Wind Nebula with Possible TeV Gamma-Ray Counterpart</i>	

TEACHING EXPERIENCE

Sailing Instructor , UC Berkeley Recreation and Wellness	April 2025 – Present
Teacher , <i>Python4Physics</i> public coding course	Summer 2024, 2025
UC Berkeley <i>SPLASH</i> Teacher (9th–12th grade)	April 2020, 2021, 2022
<i>24 Hours in the Sky</i>	
<i>From Stars to Gold</i>	
Teaching Assistant , Physics Department, Barnard College	January 2020 – May 2020
<i>Lab instructor for Electricity & Magnetism</i>	
Staff, Physics & Astronomy Helproom , Barnard College	January 2018 – December 2019
<i>Tutored Mechanics, Electromagnetism, and Waves and Optics</i>	
Peer-to-Peer Tutor , Barnard College	January 2017 – May 2018
<i>Small-group tutoring for peers in the Calculus sequence and Introductory Astronomy</i>	

STUDENTS MENTORED

Esha Rao (UC Berkeley Graduate Student)	May 2026 – Present
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SERVICE

Graduate Student Representative, Faculty Mentoring Committee, UC Berkeley Physics	Fall 2022, Spring 2023
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EQUITY & OUTREACH

Astro on Tap Public Talk, Oakland, CA, <i>Hot Big Bang, Ice Cold Scientists</i>	July 2026
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Facilitator, Respect is a Part of Research Department Training, UC Berkeley Physics August 2022 – August 2025

REYES (Remote Experience for Young Engineers and Scientists) Speaker, *Astrophysics from the South Pole* July 2025

Founder & Organizer, Physics@OUSD (speaker series for Oakland public high school physics classes) August 2023 – May 2025

Mentor in SWPS, POWER, and PODS programs for gender minorities in physics 2022-2025

PUBLICATIONS (CONT.)

Archipley, M., et al. (incl. **A. Coerver**). “Millimeter-wave observations of Euclid Deep Field South using the South Pole Telescope: A data release of temperature maps and catalogs”. *Astronomy & Astrophysics* (2026). DOI: [10.1051/0004-6361/202555798](https://doi.org/10.1051/0004-6361/202555798). arXiv: [2506.00298](https://arxiv.org/abs/2506.00298) [[astro-ph.GA](#)].

Athreya, H., et al. (incl. **A. Coerver**). *The SPT-3G+ receiver design*. 2026. DOI: [10.48550/arXiv.2608.20235](https://doi.org/10.48550/arXiv.2608.20235). arXiv: [2608.20235](https://arxiv.org/abs/2608.20235) [[astro-ph.IM](#)].

Bleem, L. E., et al. (incl. **A. Coerver**). *Galaxy Clusters Selected via the Sunyaev-Zel’dovich Effect in 5 year data from the SPT-3G Main Survey*. 2026. DOI: [10.48550/arXiv.2607.01175](https://doi.org/10.48550/arXiv.2607.01175). arXiv: [2607.01175](https://arxiv.org/abs/2607.01175) [[astro-ph.CO](#)].

Camphuis, E., et al. (incl. **A. Coerver**). “SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G main field”. *Physical Review D* (2026). DOI: [10.1103/7wt3-9v2y](https://doi.org/10.1103/7wt3-9v2y). arXiv: [2506.20707](https://arxiv.org/abs/2506.20707) [[astro-ph.CO](#)].

Chaubal, P., et al. (incl. **A. Coerver**). *SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power*. 2026. DOI: [10.48550/arXiv.2601.20551](https://doi.org/10.48550/arXiv.2601.20551). arXiv: [2601.20551](https://arxiv.org/abs/2601.20551) [[astro-ph.CO](#)].

de Haan, Tijmen, et al. (incl. **A. Coerver**). “Characterization of the Polarization Beam Response of SPT-3G Using Point Sources”. *The Open Journal of Astrophysics* (2026). DOI: [10.33232/001c.165326](https://doi.org/10.33232/001c.165326). arXiv: [2602.06334](https://arxiv.org/abs/2602.06334) [[astro-ph.CO](#)].

Khalife, A. R., et al. (incl. **A. Coerver**). “SPT-3G D1: Axion early dark energy with CMB experiments and DESI observations”. *Physical Review D* (2026). DOI: [10.1103/8jjr-7hpb](https://doi.org/10.1103/8jjr-7hpb). arXiv: [2507.23355](https://arxiv.org/abs/2507.23355) [[astro-ph.CO](#)].

Kornoelje, K., et al. (incl. **A. Coerver**). “The SPT-deep Cluster Catalog: Sunyaev-Zel’dovich Selected Clusters from Combined SPT-3G and SPTpol Measurements over 100 Square Degrees”. *The Astrophysical Journal* (2026). DOI: [10.3847/1538-4357/ae5f7b](https://doi.org/10.3847/1538-4357/ae5f7b). arXiv: [2503.17271](https://arxiv.org/abs/2503.17271) [[astro-ph.CO](#)].

Levy, K., et al. (incl. **A. Coerver**). *CMB Lensing Reconstruction Using Two Years of Temperature Data from the SPT-3G Summer Survey*. 2026. DOI: [10.48550/arXiv.2607.05784](https://doi.org/10.48550/arXiv.2607.05784). arXiv: [2607.05784](https://arxiv.org/abs/2607.05784) [[astro-ph.CO](#)].

Maniyar, A. S., et al. (incl. **A. Coerver**). “SPT-3G D1: Compton-y maps using data from the SPT-3G and Planck surveys”. *Physical Review D* (2026). DOI: [10.1103/7mdh-jpd3](https://doi.org/10.1103/7mdh-jpd3). arXiv: [2602.11279](https://arxiv.org/abs/2602.11279) [[astro-ph.CO](#)].

- Nakato, Y., et al. (incl. **A. Coerver**). *SPT-3G D1: Foreground-Robust Lensing Templates for Primordial Gravitational Wave Searches*. 2026. DOI: [10.48550/arXiv.2608.06343](https://doi.org/10.48550/arXiv.2608.06343). arXiv: [2608.06343](https://arxiv.org/abs/2608.06343) [astro-ph.CO].
- Omori, Y., et al. (incl. **A. Coerver**). *SPT-3G D1: Quadratic-Estimator CMB Lensing Reconstruction and Cosmology*. 2026. DOI: [10.48550/arXiv.2608.31136](https://doi.org/10.48550/arXiv.2608.31136). arXiv: [2608.31136](https://arxiv.org/abs/2608.31136) [astro-ph.CO].
- Ouellette, A., et al. (incl. **A. Coerver**). *Cross-correlation of SPT-3G D1 CMB lensing and DES Y3 galaxy lensing*. 2026. DOI: [10.48550/arXiv.2606.26223](https://doi.org/10.48550/arXiv.2606.26223). arXiv: [2606.26223](https://arxiv.org/abs/2606.26223) [astro-ph.CO].
- Qu, Frank J., et al. (incl. **A. Coerver**). “Unified and Consistent Structure Growth Measurements from Joint ACT, SPT, and Planck CMB Lensing”. *Physical Review Letters* (2026). DOI: [10.1103/k5yr-3h6d](https://doi.org/10.1103/k5yr-3h6d). arXiv: [2504.20038](https://arxiv.org/abs/2504.20038) [astro-ph.CO].
- Quan, W., et al. (incl. **A. Coerver**). *SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field*. 2026. DOI: [10.48550/arXiv.2603.20163](https://doi.org/10.48550/arXiv.2603.20163). arXiv: [2603.20163](https://arxiv.org/abs/2603.20163) [astro-ph.CO].
- Raghunathan, S., et al. (incl. **A. Coerver**). “Measurement of the Full Shape of the Thermal Sunyaev-Zel’dovich Power Spectrum from the South Pole Telescope and Herschel-SPIRE Observations”. *The Astrophysical Journal Letters* (2026). DOI: [10.3847/2041-8213/ae5c07](https://doi.org/10.3847/2041-8213/ae5c07). arXiv: [2602.10107](https://arxiv.org/abs/2602.10107) [astro-ph.CO].
- Simpson, A., et al. (incl. **A. Coerver**). *Polarization Angle Swings in Blazars Detected in the Millimeter-wave with the South Pole Telescope*. 2026. DOI: [10.48550/arXiv.2608.00302](https://doi.org/10.48550/arXiv.2608.00302). arXiv: [2608.00302](https://arxiv.org/abs/2608.00302) [astro-ph.HE].
- Tandoi, C., et al. (incl. **A. Coerver**). *Millimeter-wave Detections of Symbiotic Stars in SPT and ACT Data*. 2026. DOI: [10.48550/arXiv.2605.01022](https://doi.org/10.48550/arXiv.2605.01022). arXiv: [2605.01022](https://arxiv.org/abs/2605.01022) [astro-ph.SR].
- Vitrier, A., et al. (incl. **A. Coerver**). “Towards constraining cosmological parameters with SPT-3G observations of 25% of the sky”. *Physical Review D* (2026). DOI: [10.1103/8yfb-t1j9](https://doi.org/10.1103/8yfb-t1j9). arXiv: [2510.24669](https://arxiv.org/abs/2510.24669) [astro-ph.CO].
- Wan, Y., et al. (incl. **A. Coerver**). “Detection of Millimeter-wavelength Flares from Two Accreting White Dwarf Systems in the SPT-3G Galactic Plane Survey”. *The Astrophysical Journal* (2026). DOI: [10.3847/1538-4357/ae2de8](https://doi.org/10.3847/1538-4357/ae2de8). arXiv: [2509.08962](https://arxiv.org/abs/2509.08962) [astro-ph.SR].
- L. Balkenhol, **Coerver, A.**, C. L. Reichardt, and J. A. Zebrowski. “Probing Anisotropic Cosmic Birefringence with Foreground-Marginalised SPT B-mode Likelihoods”. *The Open Journal of Astrophysics* (2025). DOI: [10.33232/001c.147459](https://doi.org/10.33232/001c.147459). arXiv: [2510.07928](https://arxiv.org/abs/2510.07928) [astro-ph.CO].
- Chichura, P. M., et al. (incl. **A. Coerver**). “Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning”. *Journal of Astronomical Instrumentation* (2025). DOI: [10.1142/S2251171725500011](https://doi.org/10.1142/S2251171725500011). arXiv: [2412.15167](https://arxiv.org/abs/2412.15167) [astro-ph.IM].
- Foster, Allen, et al. (incl. **A. Coerver**). “Detection of Thermal Emission at Millimeter Wavelengths from Low-Earth Orbit Satellites”. *The Open Journal of Astrophysics* (2025). DOI: [10.33232/001c.137526](https://doi.org/10.33232/001c.137526). arXiv: [2411.03374](https://arxiv.org/abs/2411.03374) [astro-ph.IM].
- Ge, F., et al. (incl. **A. Coerver**). “Cosmology from CMB lensing and delensed EE power spectra using 2019–2020 SPT-3G polarization data”. *Physical Review D* (2025). DOI: [10.1103/PhysRevD.111.083534](https://doi.org/10.1103/PhysRevD.111.083534). arXiv: [2411.06000](https://arxiv.org/abs/2411.06000) [astro-ph.CO].
- Ansarinejad, B., et al. (incl. **A. Coerver**). “Mass calibration of DES Year-3 clusters via SPT-3G CMB cluster lensing”. *Journal of Cosmology and Astroparticle Physics* (2024). DOI: [10.1088/1475-7516/2024/07/024](https://doi.org/10.1088/1475-7516/2024/07/024). arXiv: [2404.02153](https://arxiv.org/abs/2404.02153) [astro-ph.CO].

- Prabhu, K., et al. (incl. **A. Coerver**). “Testing the Λ CDM Cosmological Model with Forthcoming Measurements of the Cosmic Microwave Background with SPT-3G”. *The Astrophysical Journal* (2024). DOI: [10.3847/1538-4357/ad5ff1](https://doi.org/10.3847/1538-4357/ad5ff1). arXiv: [2403.17925](https://arxiv.org/abs/2403.17925) [[astro-ph.CO](#)].
- Raghunathan, S., et al. (incl. **A. Coerver**). “First Constraints on the Epoch of Reionization Using the Non-Gaussianity of the Kinematic Sunyaev-Zel’dovich Effect from the South Pole Telescope and Herschel-SPIRE Observations”. *Physical Review Letters* (2024). DOI: [10.1103/PhysRevLett.133.121004](https://doi.org/10.1103/PhysRevLett.133.121004). arXiv: [2403.02337](https://arxiv.org/abs/2403.02337) [[astro-ph.CO](#)].
- Tandoi, C., et al. (incl. **A. Coerver**). “Flaring Stars in a Nontargeted Millimeter-wave Survey with SPT-3G”. *The Astrophysical Journal* (2024). DOI: [10.3847/1538-4357/ad58db](https://doi.org/10.3847/1538-4357/ad58db). arXiv: [2401.13525](https://arxiv.org/abs/2401.13525) [[astro-ph.SR](#)].
- Balkenhol, L., et al. (incl. **A. Coerver**). “Measurement of the CMB temperature power spectrum and constraints on cosmology from the SPT-3G 2018 TT, TE, and EE dataset”. *Physical Review D* (2023). DOI: [10.1103/PhysRevD.108.023510](https://doi.org/10.1103/PhysRevD.108.023510). arXiv: [2212.05642](https://arxiv.org/abs/2212.05642) [[astro-ph.CO](#)].
- Pan, Z., et al. (incl. **A. Coerver**). “Measurement of gravitational lensing of the cosmic microwave background using SPT-3G 2018 data”. *Physical Review D* (2023). DOI: [10.1103/PhysRevD.108.122005](https://doi.org/10.1103/PhysRevD.108.122005). arXiv: [2308.11608](https://arxiv.org/abs/2308.11608) [[astro-ph.CO](#)].
- M. S. R. Freeman, J. R. Lu, J. Lyke, J. Delorme, E. Ramey, G. Jung, S. Cetre, S. Lilley, P. Richards, E. Wetherell, A. Surendran, **Coerver, A.**, and P. Wizinowich. “An optical distortion solution for the Keck I OSIRIS Imager”. *Adaptive Optics Systems VIII*. 2022. DOI: [10.1117/12.2630712](https://doi.org/10.1117/12.2630712).
- K. Mori, C. J. Hailey, S. Mandel, T. Schutt, M. Bachetti, **Coerver, A.**, F. K. Baganoff, H. Dykaar, J. E. Grindlay, D. Haggard, K. Heuer, J. Hong, B. J. Hord, C. Jin, M. Nynka, G. Ponti, and J. A. Tomsick. “NuSTAR and Chandra Observations of New X-Ray Transients in the Central Parsec of the Galaxy”. *The Astrophysical Journal* (2019). DOI: [10.3847/1538-4357/ab4b47](https://doi.org/10.3847/1538-4357/ab4b47). arXiv: [1910.03459](https://arxiv.org/abs/1910.03459) [[astro-ph.HE](#)].